

Implementing a RAG-Based Chatbot in Industrial Operations: A Real-World Case Study

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Abstract—Retrieval-Augmented Generation (RAG) has gained significant popularity due to its ability to ground generative models in specific knowledge sources. It can be especially beneficial in high-speed, safety-critical manufacturing settings, given its capacity to reduce information retrieval time, enhance response accuracy, and minimize the risk of hallucinated answers. Given the recency of RAG as a technological innovation, practical implementations are especially valuable, as they offer evidence of its adaptability to specific domains. In this paper, we investigate the real-world application of a RAG chatbot within an industrial context. Using procedural and quality-related documentation from a fast-moving consumer goods (FMCG) manufacturing plant, supported by expert-annotated reference answers, we experiment with various model configurations. Our deployed pipeline achieves faithfulness of 92.63%, answer relevancy of 91.31%, context precision of 75.20%, and context recall of 86.60%. The configuration and codebase for the proposed framework is available on Github.

Index Terms—question answering, retrieval augmented generation (RAG), Large Language Models (LLMs), GPT

I. INTRODUCTION

Retrieval-Augmented Generation (RAG) has emerged as a widely adopted approach for grounding generative language models in domain-specific knowledge sources. Its use is particularly advantageous in fast-paced, safety-critical manufacturing environments, where it can significantly accelerate information access, improve answer reliability, and greatly reduce the likelihood of generating hallucinated responses [1]. As RAG is a relatively recent advancement, real-world implementations are instrumental in understanding its effectiveness, limitations, and domain-specific adaptability.

In this study, we explore the real-world application of a RAG chatbot within an industrial setting. Our goal is to assess how such systems can support day-to-day operations by improving access to critical procedural knowledge. We develop an RAG pipeline tailored for a plant operated by a Fast-Moving Consumer Goods (FMCG) company. The system is trained on internal procedural documents, and we conduct extensive experiments across various configurations, including different Large Language Models (LLMs), embedding models, and context sizes. For evaluation, we use a domain-specific question-answer dataset, with both queries and reference answers carefully crafted by operational experts.

Upon deployment, we managed to achieve a faithfulness of 92.63%, answer relevancy of 91.31%, context precision of

75.20%, and context recall of 86.60%. Our findings suggest that RAG chatbots can offer substantial efficiency gains by significantly reducing the time required to locate information in high-pressure, time-sensitive environments typical of industrial operations. Furthermore, by configuring the RAG to cite its responses with references to source documents, we were able to enhance trust and acceptance among end users.

II. RELATED WORKS

Question answering systems are designed to fulfill human information needs [2]. In the early stages, these needs were mainly factual, such as retrieving names, definitions, etc., and were addressed using rule-based or information retrieval-based systems that relied on keyword matching and pattern templates [3]–[5]. As user expectations evolved, a demand for more structured, precise, and explainable answers, lead to the development of knowledge base question answering systems [6], [7]. With the rise of deep learning and large-scale datasets, neural reading comprehension models emerged, enabling frameworks to extract answers from unstructured text [8]–[10]. However, as real-world questions becomes increasingly open-ended and complex, these aforementioned approaches struggle with scalability, grounding, and accuracy.

In 2020, the RAG model was introduced as a hybrid approach that combines retrieval-based and generative question answering [11]. This hybrid approach was able to largely address the modern needs for question answering. At a high level, RAG consists of two primary components: a retriever and a generator. First, a pre-trained embedding model is used to encode a large corpus of documents into dense vector representations, referred to as “*indexes*”. Then, the same embedding model is employed to encode the user query into a dense vector, and a Maximum Inner Product Search [12] is performed over the index to retrieve the top K most relevant indexes. These retrieved documents are then provided as context to a sequence-to-sequence language model, which generates a natural language response based on retrieved information. The objective of RAG is to compute the distribution of the output (y) given the query (x) based on K retrieved contexts (c) with the following formula:

$$p(y | x) = \sum_{k=1}^K p(c_k | x) \cdot p(y | x, c_k) \quad (1)$$

Shortly after the introduction of RAG, the framework was extended to support multimodal inputs, enabling it to process images and tables alongside text input [13], [14]. Given RAG's effectiveness in LLMs for question answering while mitigating hallucinations and factual inaccuracies [15], its adoption has rapidly expanded across a wide range of domains. In a relatively short period, RAG-based systems have been successfully applied in diverse fields such as biology and medicine [16]–[18], customer service and support [19], [20], and law [21], [22]. As a very recent technological development, real-world use cases of RAG remain highly valuable, offering practical insights into its capabilities, limitations, and impact across different domains.

III. METHODOLOGY

A. Data

The dataset consists of 8 PDF documents, collectively containing over 178,000 words. These documents provide comprehensive information on production and quality processes and are written in Turkish. The content spans detailed textual descriptions, as well as various charts and flowcharts that represent workflows, procedures, and KPIs.

For model evaluation, 2 separate question-answer sets written in Turkish are used. These questions were curated and answered by domain experts to capture critical information from the source documents. For hyperparameter tuning, the first question-answer set comprising 128 questions and approximately 23,650 words is employed. To assess the performance of the deployed model, the second question-answer set containing 98 questions and a total of 12,960 words was used. These evaluation sets serve to measure the chatbot's ability to generate accurate and contextually grounded responses.

B. Architecture

The RAG chatbot architecture consists of 3 steps, *Indexing*, *Retrieval*, and *Augmentation*; and is detailed in Figure 1.

Step 1: Indexing. The source documents are split into overlapping text chunks of 60,000 characters, with an overlap of 1,000 characters between consecutive chunks. The chunk size is chosen to balance contextual completeness with computational efficiency. The overlap ensures that key contextual information is preserved across chunk boundaries, minimizing the risk of semantic fragmentation. Since the source documents typically contain detailed and lengthy passages, a large overlap between text chunks is maintained to match the structure of source data. No additional preprocessing is applied to the raw text. Each text chunk is embedded using a pre-trained embedding model. The resulting embeddings are stored in a vector database to be used in the next step.

Step 2: Retrieval. To respond to a user query, the query is first vectorized using the same embedding model employed in the indexing phase. This ensures consistency in the vector space used for both document chunks and queries. The resulting query vector is used to retrieve the top K most similar contexts from the vector database. Retrieval is performed using the Hierarchical Navigable Small World [23] algorithm, with

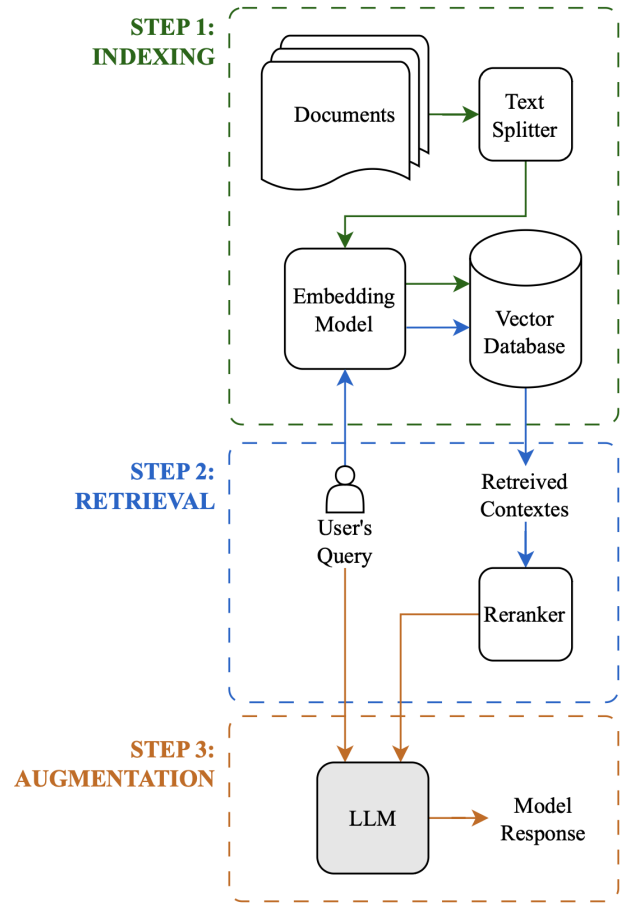


Fig. 1. Flowchart of the proposed RAG architecture

the configuration parameters selected as 4 bi-directional link count, 500 entry factor for construction, 600 entity factor for search, and cosine similarity for the similarity metric. To further improve precision, the retrieved K contexts are subsequently re-ranked using the BM-25 algorithm [24].

Step 3: Generation. In the final step, the original user query and the re-ranked K' contexts are passed to an LLM to generate the final response. The LLM is configured with a temperature of 0 to ensure and fact-focused outputs, and a maximum token limit of 1,200 to balance completeness with response length constraints. A precise system prompt is provided to the LLM, defining the chatbot's role and objectives. This includes instructions to provide citations referencing the source documents and to structure responses using steps or bullet points if appropriate to enhance clarity.

C. Performance Metrics

RAG performance is measured using faithfulness, context precision, recall, and answer relevancy.

Faithfulness is a metric that measures the proportion of statements in a model-generated response that are supported by the retrieved context, thereby quantifying the extent to which

the response is grounded in the source documents [25]:

$$\text{Faithfulness} = \frac{|S_C^M|}{|S^M|} \quad (2)$$

where S^M denotes the statements in the model-generated response, C denotes the retrieved contexts, and S_C^M denotes the statements in the model-generated response supported by the retrieved contexts. A faithfulness score closer to 1 indicates a higher degree of alignment between the response and the supporting context. This metric is especially critical for this study, as it directly evaluates the extent to which the response is grounded in the source documents.

Context Precision measures the proportion of contexts relevant to the user query, and is calculated as follows [25]:

$$\text{Context Precision@K} = \frac{\sum_{k=1}^K (\text{Precision@k} \cdot v_k)}{|R_K|} \quad (3)$$

$$\text{Precision@k} = \frac{\text{TP@k}}{\text{TP@k} + \text{FP@k}} \quad (4)$$

where K is the total number of contexts in retrieved and re-ranked contexts, $v_k \in \{0, 1\}$ is the relevance indicator at rank k , R_K is the contexts in the top K relevant to the query. TP denotes the true positive, the proportion of contexts retrieved that are in fact relevant to the query, and FP denotes the false positive, the contexts that are retrieved but are not relevant to the query. In Equations 3 and 4, two items are deemed relevant when the cosine similarity between their corresponding embeddings is greater than 0.5.

Context Recall measures how well the relevant contexts are retrieved. It is calculated as the following [25]:

$$\text{Context Recall} = \frac{|S_C^R|}{|S^R|} \quad (5)$$

where S^R denotes the statements in the reference answer created by the experts as part of the evaluation dataset, and S_C^R denotes the statements in the reference answer supported by the retrieved contexts.

Answer Relevancy quantifies the relevancy between the user query and the model-generated response. However, it is not directly calculated by computing the similarity between the query and the response as they tend to be structurally dissimilar, and therefore their similarity is typically misleading. Instead, the metric is computed using similarity between structurally and semantically comparable text. Firstly, for every user query, 3 questions are generated by the model. Then, the cosine similarity between the embeddings of these model-generated questions and the user query is averaged to calculate the answer relevancy with the following equation [25]:

$$\text{Answer Relevancy} = \frac{1}{3} \sum_{i=1}^3 \frac{E_{G_i} \cdot E_Q}{\|E_{G_i}\| \|E_Q\|} \quad (6)$$

where E_{G_i} denotes the embedding of the i^{th} model-generated question and E_Q denotes the embedding of the query.

Figure 2 illustrates how the performance metrics relate to the key components; query, response, context, and ground truth answer.

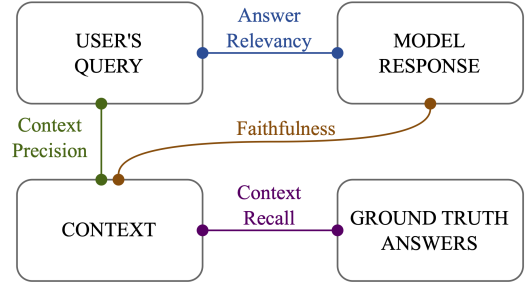


Fig. 2. Performance metrics in relation to query, response, context, and ground truth answer. Adapted from [26].

D. Component-Wise Model Selection and Validation

Using the question-answer dataset for tuning, we experiment with various model configurations. These configurations are evaluated via context precision, answer relevancy, faithfulness, and context recall, using the RAGAS library [25]. The tuning is performed in 3 consecutive steps, to select the LLM, the embedding model, and the number of contexts input to the LLM, respectively.

(i) Firstly, using the text-embedding-ada-002 model, setting the number of contexts retrieved by the similarity search (K) and the number of contexts returned by the reranker (K') to 5, we deploy the architecture detailed in Table I using LLMs GPT-4o, GPT-4.1 nano, and GPT-4o mini [27]. Among these, GPT-4.1 nano achieves the highest performance, with an average score of 87.48% across 4 evaluation metrics, outperforming GPT-4o by 8.26% and GPT-4o mini by 0.75%. Notably, GPT-4.1 nano demonstrates the strongest performance in the faithfulness metric (see Table I), which is particularly important in this study, as it reflects the degree to which model outputs are grounded in the source documents. Although GPT-4o mini also exhibits competitive performance, GPT-4.1 nano is ultimately selected due to its superior accuracy and approximately 66% lower cost.

(ii) Secondly, fixing the LLM to GPT-4.1 nano, and setting both the number of contexts retrieved by the similarity search (K) and the number of contexts returned by the reranker (K') to 5, we experiment with 3 embedding models: text-embedding-ada-002, text-embedding-3-large, and text-embedding-3-small [27]. text-embedding-ada-002 achieves the highest average performance of 87.48%, surpassing text-embedding-3-large and text-embedding-3-small by 1.59% and 3.10%, respectively. Based on these results, text-embedding-ada-002 is selected for the final system configuration.

(iii) Lastly, we investigate the impact of varying the number of contexts retrieved by the similarity search (K) and the number of contexts returned by the reranker (K'). As illustrated in Figure 1, several configurations are compared. We observe that in cases ($K=10, K'=5$ vs. $K=K'=5$), ($K=10, K'=3$ vs. $K=K'=3$), and ($K=5, K'=3$ vs. $K=K'=3$), the latter always performs better despite the number of contexts ultimately input to the LLM remain unchanged. This suggests that the reranker should not filter out retrieved contexts, i.e., K' should

TABLE I
MODEL SELECTION AND VALIDATION OUTPUTS

Embedding Model	K	K'	LLM	Faithfulness	Answer Relevancy	Context Recall	Context Precision
text-embedding-ada-002	5	5	GPT-4o	0.8559	0.8558	0.7692	0.6880
text-embedding-ada-002	5	5	GPT-4.1 nano	0.9041	0.9048	0.8783	0.8119
text-embedding-ada-002	5	5	GPT-4o mini	0.8769	0.8598	0.8242	0.9083
text-embedding-ada-002	5	5	GPT-4.1 nano	0.9041	0.9048	0.8783	0.8119
text-embedding-3-large	5	5	GPT-4.1 nano	0.8955	0.8189	0.8836	0.8377
text-embedding-3-small	5	5	GPT-4.1 nano	0.9060	0.7984	0.8684	0.8042
text-embedding-ada-002	10	10	GPT-4.1 nano	0.9263	0.9131	0.8660	0.7520
text-embedding-ada-002	5	5	GPT-4.1 nano	0.9041	0.9048	0.8783	0.8119
text-embedding-ada-002	3	3	GPT-4.1 nano	0.8871	0.8844	0.8610	0.8402
text-embedding-ada-002	10	5	GPT-4.1 nano	0.8809	0.8844	0.8554	0.7919
text-embedding-ada-002	10	3	GPT-4.1 nano	0.8157	0.8097	0.8097	0.8097
text-embedding-ada-002	5	3	GPT-4.1 nano	0.8675	0.8582	0.8575	0.8268

not be smaller than K . Based on the evaluation across all four metrics with a particular emphasis on faithfulness, we select $K = K' = 10$ for the final system configuration.

IV. REAL-WORLD DEPLOYMENT

A. System Configuration

The model is deployed for real-world production use via Microsoft Azure, Amazon Elastic Compute Cloud (EC2), and a custom web interface.

Step 1: The source documents are stored in Microsoft Azure Blob Storage, which serves as the main data repository. A custom text-splitting script is executed within a private Amazon EC2 container to split the documents into overlapping chunks. These processed chunks are then uploaded to a separate Azure Blob Storage container for subsequent use. The embedding generation and vector storage are handled via Microsoft Azure AI Search. Within this service, the embedding model is integrated into the “*indexer*” component, the vector database is stored under “*index*”, and the configurations related to the embedding model and the vector database are managed via the “*skillset*”.

Step 2: Users submit queries through a custom-built web interface. These queries are managed by Microsoft Azure Machine Learning Service. The Machine Learning Service sends the queries to the Microsoft Azure AI Search Service to have their embeddings generated, the relevant contexts retrieved from the vector database and re-ranked.

Step 3: This step is again managed by Microsoft Azure Machine Learning Service. The Machine Learning Service passes original user queries and the re-ranked contexts to the LLM, and returns the LLM-generated response.

Additionally, the system is configured to check for updates to the source documents on a daily basis. Newly detected documents are processed through the same pipeline (splitting and indexing). This continuous update cycle is particularly vital in the fast-paced FMCG industry, where operational protocols and product information are subject to frequent changes. Maintaining up-to-date embeddings ensures that the chatbot provides accurate, relevant, and timely information.

B. Usage Statistics and Sample Interactions

The deployed model processes an average of 4 requests per minute, with a latency of approximately 4,000 ms per request. Table II presents sample user queries alongside the model-generated responses and expert-verified reference answers.

C. Performance Evaluation and Key Findings

The deployed model is evaluated using the held-out question-answer set designated for this testing phase. The final pipeline achieves a faithfulness of 92.63%, answer relevancy of 91.31%, context precision of 75.20%, and context recall of 86.60%. When compared to the best configuration obtained during the tuning phase, which yielded 90.41% faithfulness, 90.48% answer relevancy, 81.19% context precision, and 87.83% context recall, no drop in our key performance metric faithfulness is observed. This suggests that the architecture generalizes well and that its performance with respect to groundedness is robust to future updates in the source documents and content.

We observed that the model-generated responses are generally longer than the expert-provided answers. While it is possible to simplify and shorten these outputs, we deliberately chose to retain their extended form. This decision is based on the hypothesis that the experts, being highly knowledgeable, may omit explanations for commonly understood terms or concepts. In contrast, the deployed RAG system is intended to support employees of varying expertise levels. Therefore, we prioritized clarity and completeness over strict alignment with the reference answers. Although this choice affects performance metrics due to embedding differences caused by length, we consider detailed, well-explained responses more appropriate for the intended use case.

We configured the LLM to cite its answers by referencing the source documents from which the information was retrieved. This simple yet effective addition significantly improved user trust in the RAG tool and led to increased adoption across the organization. For initial deployments of similar AI systems, we strongly recommend incorporating transparent citation mechanisms, as they not only enhance interpretability but also boost user confidence in the system’s outputs.

TABLE II
SAMPLE QUERIES, MODEL RESPONSES AND REFERENCE ANSWERS

Query	Model Response	Reference Answer
DISC yaklaşımı nedir? (What is the DISC approach?)	DISC yaklaşımı, çalışma sırasında kullanılan bir kişilik envanteridir. Bu yaklaşım, bireylerin farklı renk enerjilerini taşıdığını varsayar ve her bireyin farklı özellikler sergilediğini kabul eder. DISC yaklaşımına göre, herkes farklı renk enerjilerini taşır ve en ideali, her profilin bu renk enerjilerinin dengeli bir şekilde taşınmasıdır. Ayrıca, profillerde İyi-Kötü, Doğru-Yanlış gibi tanımlamaların doğru olmadığı belirtilir. Bu yaklaşım, mantıksal, planlı ve sonuç odaklı kişilik özelliklerini içerir. Kaynak: D₁ dokümanı (The DISC approach is a personality inventory used during work. This approach assumes that individuals carry different color energies and accepts that each individual exhibits different characteristics. According to the DISC approach, everyone carries different color energies, and the ideal is for each profile to carry these color energies in a balanced manner. It is also stated that definitions such as Good-Bad or Right-Wrong are not accurate in the context of profiles. This approach includes logical, planned, and result-oriented personality traits. Source: Document D₁)	DISC, çalışma sırasında kullanılan bir kişilik envanteridir. DISC yaklaşımına göre herkes farklı renk enerjilerini taşır. En ideali her profili eşit şekilde taşımak ve dengeli olmaktır. Profillerde İyi-Kötü, Doğru-Yanlış gibi tanımlamalar doğru değildir. (DISC is a personality inventory used during work. According to the DISC approach, everyone carries different color energies. The ideal is to carry each profile equally and to be balanced. Definitions such as Good-Bad or Right-Wrong are not accurate for the profiles.)
Hat Performans Sistemi nedir ve nasıl çalışır? (What is the Line Performance System and how does it work?)	Hat Performans Sistemi, üretim hattının verimliliğini ve etkinliğini takip etmek amacıyla kullanılan bir sistemdir. Bu sistem, vardiya bazında tesisin planlı ve plansız kayıplarını izler ve analiz eder. Sistem kapsamında takip edilen ana kalemler; tesisin etkin çalıştığı süre, temizlik süreleri, imalat dönüşüm süreleri, ıskarta ürünler, GSF (Gelişmiş Sistem Farkları), duruş süreleri ve planlı bakım çalışmalarıdır. Kaynak:D₂ dokümanı (The Line Performance System is a system used to monitor the efficiency and effectiveness of the production line. This system tracks and analyzes the facility's planned and unplanned losses on a shift basis. The main items monitored within the scope of the system include: the facility's effective operating time, cleaning durations, manufacturing conversion times, scrap products, GSF (Advanced System Differences), downtime durations, and planned maintenance activities. Source: Document D₂)	Hat Performans Sistemi, hatta yapılan üretimin verimliliğini, tesisin planlı ve plansız kayıplarını vardiya bazında takip eden bir sistemdir. Hat performansında takip edilen ana kalemler; tesisin etkin çalıştığı süre, temizlik süreleri, imalat dönüşüm süreleri, ıskarta, GSF, duruş süreleri ve planlı bakım çalışmalarıdır. (The Line Performance System is a system that monitors the efficiency of production on the line and the facility's planned and unplanned losses on a shift basis. The main items tracked in line performance include: the facility's effective operating time, cleaning durations, manufacturing conversion times, scrap, GSF, downtime durations, and planned maintenance activities.)
16 Büyük Kayıp nedir ve neden önemlidir? (What are the 16 Major Losses and why are they important?)	16 Büyük Kayıp, fabrikanın kayıp yapısı analizinde belirlenen en büyük kayıplar kategorisidir. Bu kayıplar, işletmenin toplam kayıplarının önemli bir kısmını oluşturur ve para birimi cinsinden belirtilir. Bu analiz, kayıpların oran veya miktar değil, maliyet açısından sıralanmasını sağlar. Bu sayede, en büyük kayıplara odaklanılarak maliyetlerin azaltılması ve verimliliğin artırılması hedeflenir. Önemlidir çünkü, Kayıp Yapısı Analizi, işletmenin bütünsel bir bakış açısıyla kayıplarını detaylı kırılımlarla ortaya koyar ve odaklanılması gereken alanları netleştirir. Bu sayede, kaynakların etkin kullanımı sağlanır ve maliyetlerin düşürülmesi ile işletmenin rekabet gücü artar. Ayrıca, bu analiz, iyileştirme çalışmalarının temelini oluşturarak, işletmenin performansını ve karlılığını artırmaya yardımcı olur. Kaynak: D₃ dokümanı (The 16 Major Losses are the largest loss categories identified in the factory's Loss Structure Analysis. These losses constitute a significant portion of the total losses of the operation and are expressed in monetary terms. This analysis enables the ranking of losses based on cost rather than ratio or quantity. In this way, the aim is to reduce costs and increase efficiency by focusing on the largest losses. They are important because the Loss Structure Analysis presents the operation's losses with detailed breakdowns from a holistic perspective and clarifies the areas that need focus. This ensures the efficient use of resources and increases the company's competitiveness by reducing costs. Moreover, this analysis forms the basis for improvement efforts and helps enhance the company's performance and profitability. Source: Document D₃)	Operasyonel Mükemmellik çalışmaları; üretim sahasındaki kayıpların tümüyle yok edilmesini ve verimliliği maksimum olan bir üretim sistemi yaratmayı hedeflemektedir. Bunun için kayıpları bilmek, ölçmek, çözüm üretmek ve bu çözümleri uygulamak gerekmektedir. Kayıplarla mücadele etmek için öncelikle kayıpların nerelerde olduğu ve neden olduğu tespit edilmelidir. Bu tespitlerin sistematik yapılması için kayıplar aşağıdaki sınıflandırmaya göre değerlendirilir. Kayıpların tanımlanması, kayıplarla ilgili farkındalığın artırılması, kök nedenlerinin bulunması, iyileştirmelerin hayata geçirilmesi ve kazanımların sürdürülmesine odaklanılır. Kullandığımız araç ve sistemler kayıplarımızın ortadan kaldırılmasına hizmet etmektedir. (Operational Excellence efforts aim to completely eliminate losses on the production floor and create a production system with maximum efficiency. For this, it is necessary to identify, measure, generate solutions for, and implement those solutions to address the losses. In order to combat losses, it is essential first to determine where and why they occur. To systematically make these identifications, losses are evaluated according to the classification below. We focus on identifying the losses, raising awareness about them, finding root causes, implementing improvements, and sustaining the gains. The tools and systems we use are aimed at eliminating our losses.)
Sıralı Toplantı Zinciri (T-3) süresi ne kadardır? (How long is the Sequential Meeting Chain (T-3)?)	Sıralı Toplantı Zinciri (T-3) süresi 15-20 dakika olarak belirlenmiştir. Sabah 09:10-09:15 arasında yapılmaktadır. Kaynak: D₄ dokümanı (The duration of the Sequential Meeting Chain (T-3) is set to be 15–20 minutes. It is held between 09:00–09:15 in the morning. Source: Document D₄)	15-20 dakika sürer ve sabah 09:00-09:15 saatleri arasında yapılır. (It lasts 15–20 minutes and is held between 09:00–09:15 in the morning.)

V. CONCLUSION

In this study, we developed a RAG chatbot tailored for industrial operations. We detailed our model selection experiments, evaluating various LLMs, embedding models, and context sizes to identify the optimal configuration. The deployed system achieved a faithfulness of 89.07%, answer relevancy of 90.87%, context precision of 90.87%, and context recall of 90.87%. Beyond these quantitative results, the chatbot has proven to be significantly beneficial in improving operational efficiency. Moreover, its use has fostered greater trust in artificial intelligence powered solutions within the organization, laying the foundation for future applications.

Observations from real-world usage revealed an emerging need: expanding the range of data sources integrated into the system. While the tool has been effective in enabling plant personnel to quickly retrieve information and perform necessary analyses or tasks, a limitation became apparent: Users must already know that the relevant information exists in the procedural source documents indexed by the system, rather than in other commonly used portals or repositories. Unfortunately, the current pipeline supports retrieval from only a single predefined data source and lacks the capability to route queries to the appropriate source based on their content.

For future work, we plan to explore the integration of agentic RAG systems. Such systems are capable of identifying the appropriate data source required to process a given query and dynamically retrieving information from that source. Moreover, this direction will enable the use of specialized embedding models tailored to different data sources and hence types, such as tabular, text, image, etc. Overall, agentic RAG applications inherently address a key limitation of current systems: the requirement for users to know in advance that the relevant information resides in the source documents of the model.

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